

Ansh Gupta

Data Scientist, Kshema General Insurance Ltd
B.Tech. : Electrical Engineering - IIT Kanpur

Email : anshgupta.ccf@gmail.com | Mobile : (+91) 735-551-0645
LinkedIn : ansh-gupta-23a879144

Academic Qualifications

Year	Educational Qualification	Institute	CGPA/%
2017 - 2021	B.Tech.	Indian Institute of Technology, Kanpur	6.9/10.0
2017	CBSE(XII)	Jawahar Navodaya Vidyalaya, Kottayam	95.4%
2015	CBSE(X)	Jawahar Navodaya Vidyalaya, Sant Kabir Nagar	10.0/10.0

Scholastic Achievements

- Secured **All India Rank 1637** in **IIT-JEE Advanced 2017** among 0.2 million candidates
- Secured **All India Rank 2276** in **IIT-JEE Mains 2017** among 1.2 million candidates
- Secured **All India Rank 570** under **Stream SX** in **KVPY 2016** conducted by **IISc Bangalore**
- Awarded a **Minor degree** in **Industrial and Management Engineering** from IIT Kanpur

Professional Experience

- Data Scientist | Kshema General Insurance Ltd, Hyderabad, India** *Feb'24-Present*
 - Designed and automated a crop insurance Claim Management System, reducing turnaround time from days to minutes
 - Represented the company at the **National Level** under the YES-TECH program of PMFBY at TNAU Coimbatore
 - Developed a state-of-the-art object detection model using YOLOv8 architecture to identify crop types from field images
- Junior Data Scientist | Innoplexus Consulting Services Pvt Ltd, Pune, India** *Sept'21-Feb'24*
 - Developed novel solutions for addressing Drug Discovery challenges by harnessing a spectrum of AI models
 - Received the **Best Performer** award for securing **Royalty Partnership** for the company through data-driven strategies and predictive modeling with an accuracy over 90%, leveraging LLMs, NLP and Computer Vision techniques
 - Collaborated with cross-functional teams to define and refine project objectives, ensuring alignment with business goals
 - Guided a team of Junior Data Scientists to build end-to-end ML pipelines for Client projects as well as internal initiatives

Key Projects

- LCB (Light Chain Biosciences) - Client Project | Innoplexus** *Apr'22-Oct'22*
 - Developed a generative AI model using fine-tuned GPT-2 to synthesize novel amino acid sequences with enhanced binding affinities for antigens, accelerating drug development and therapeutic advancements
 - Utilized CNN and Random Forest regression models to predict binding affinities and prioritize antibody sequences
 - Shared 100 optimized sequences; wet lab tests showed 80% above benchmark, 10% below but binding, and 10% not binding
- Network Simulation (Gene Protein Interaction) - Innovation Project | Innoplexus** *Apr'23-Aug'23*
 - Performed data collection, cleaning and processing, including ETL, aggregation and scraping from multiple databases
 - Trained a GCN (Graph Convolutional Network) model in semi-supervised fashion to predict cancer driver genes in a network
 - Implemented PyTorch Geometric GNN Explainer for model interpretability, providing node and edge importance masks
- CMS (Claim Management System) - Product Development Project | Kshema** *May'24-Present*
 - Developed a pixel classification model for flood mapping using Sentinel-1 satellite data to track flooded regions
 - Identified crop-damaged pixels in agricultural land clusters by analyzing pre- and post-event Sentinel-2 satellite imagery
 - Conducted time series analysis utilizing NDVI values from remote sensing data to determine optimal harvest date for crops
 - Built and deployed the GIS Surveyor application for claim management using Flask in a production environment

Technical Skills

- Programming/Scripting Languages:** Python, C, C++, HTML, CSS, JavaScript, JSON, XML, SQL, Hive
- Libraries:** Keras, OpenCV, Scikit-learn, NumPy, Pandas, Matplotlib, spaCy, Hugging Face's Transformers
- Tools & Frameworks:** TensorFlow, PyTorch, Flask, Streamlit, MLflow, Hadoop, Spark, PostgreSQL, Selenium, Git, Excel
- Cloud Technologies:** AWS EC2, Azure DevOps, Azure Blob Storage, Docker

Relevant Coursework and Specializations

Deep Learning Specialization (<i>Coursera</i>)	Image Processing (<i>IITK</i>)	Python for Everybody Specialization (<i>Coursera</i>)
Machine Learning for Signal Processing (<i>IITK</i>)	Probability and Statistics (<i>IITK</i>)	Data Structure and Algorithm (<i>IITK</i>)
Data Mining and Knowledge Discovery (<i>IITK</i>)	Modern Cryptology (<i>IITK</i>)	Fundamentals of Computing (<i>IITK</i>)

Patents Filed

- US20240153604A1** : System for Prediction of the Drug Target Landscape for Therapeutic use and a Method thereof
- US20240177870A1** : System and Method of Predicting Biological activities of Chemical compound present in a new Drug
- US20240177869A1** : Method and System for generating a plurality of Antibody sequences of a Target from framework regions
- US20240177796A1** : Method and System for generating a plurality of Antibody sequences
- US20240221863A1** : Method and System for predicting a Binding affinity of Protein structures based on Deep learning
- US20240221869A1** : Method and System for converting a Protein Data Bank file into a Two - Dimensional Numerical Matrix
- US20240221109A1** : Method and System for converting a Protein Data Bank file into a Grayscale Image array